BANGLADESH UNIVERSITY OF PROFESSIONALS

FACULTY OF SCIENCE & TECHNOLOGY

DEPT. OF COMPUTER SCIENCE & ENGINEERING (CSE)

Course Name: Machine Learning Laboratory

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_Project Proposal

"Profanity & Toxicity" Filter for Kids' Gaming Chats

Group: A5

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1. Introduction

The rapid growth of online gaming has created virtual playgrounds where millions of children interact daily. However, these spaces are often plagued by "toxic" behavior—ranging from explicit profanity to subtle cyberbullying and predatory language. Standard filters often fail because they rely on static "blacklists" that cannot keep up with gaming slang or "leetspeak" (e.g., replacing "s" with "5"). This project aims to develop an intelligent, NLP-based filtration system specifically designed for children's safety, balancing effective moderation with the fast-paced nature of gaming communication.

2. Literature Review

Early online moderation systems relied on rule-based keyword filtering, where predefined blacklists were used to block offensive words. Although computationally efficient, these systems are easily bypassed using intentional misspellings or “leetspeak,” and they often fail to capture contextual meaning [1].

With the rise of Natural Language Processing (NLP), researchers began applying machine learning models such as Naive Bayes and Logistic Regression using TF-IDF features for toxicity detection. These approaches significantly improved classification accuracy compared to static filters, particularly when trained on large datasets like the Jigsaw Toxic Comment Classification dataset [2].

More recent studies emphasize the importance of contextual understanding. Transformer-based models such as BERT have demonstrated superior performance because they generate contextual word embeddings, enabling better detection of subtle insults, sarcasm, and implicit toxicity [3]. In gaming environments specifically, toxicity is often short, aggressive, and highly contextual, requiring models that consider conversational patterns rather than isolated words [4].

Overall, the literature indicates that while traditional ML models are efficient and suitable for real-time systems, contextual deep learning models provide higher accuracy in complex gaming chat scenarios. A hybrid approach combining normalization techniques with machine learning classification can therefore offer both efficiency and robustness.

3. Problem Statement

The growth of online gaming exposes children to toxic language, while existing keyword-based moderation has major limitations such as:

1. **Lack of Context Awareness** – Traditional filters misclassify common gaming terms (e.g., “kill”, “attack”) as toxic, resulting in high false positives and poor user experience.

2. **Vulnerability to Evasion Techniques** – Users bypass filters through leetspeak, character substitutions, spacing tricks, and intentional misspellings (e.g., “h@te”, “5tup1d”). Rule-based systems fail to detect such adversarial patterns.

3. **Absence of Severity Differentiation** – Most systems use binary classification (Toxic vs. Non-Toxic), treating mild slang and severe threats equally. For children’s gaming platforms, granular classification is necessary to prioritize high-risk content.

4. Objectives:

- To implement a multi-level toxicity classification system
- To improve confidence reliability for child safety
- To analyze context sensitivity in short gaming messages

5. Tech Stack (Technical Requirements)

This section outlines the software and tools required to build, train, and deploy the ML model.

- **Programming Language:** Python 3.9+ (The industry standard for NLP and ML).
- **Natural Language Processing (NLP) Libraries:**

* **NLTK / spaCy:** Used for tokenization, lemmatization, and removing stop-words.

    - **SciSpacy (Optional):** If specialized terminology is required.
- **Machine Learning Framework:**

  * **Scikit-learn:** For implementing TF-IDF Vectorization and classifiers like Naive Bayes, Logistic Regression, or Random Forest.

- **Data Manipulation & Visualization:**

  * **Pandas & NumPy:** For cleaning and structuring the dataset.

    - **Matplotlib & Seaborn:** For plotting the confusion matrix and accuracy curves.
- **Deployment/UI:**

  * **Streamlit:** To create a front-end dashboard where users can input text and see real-time toxicity scores.

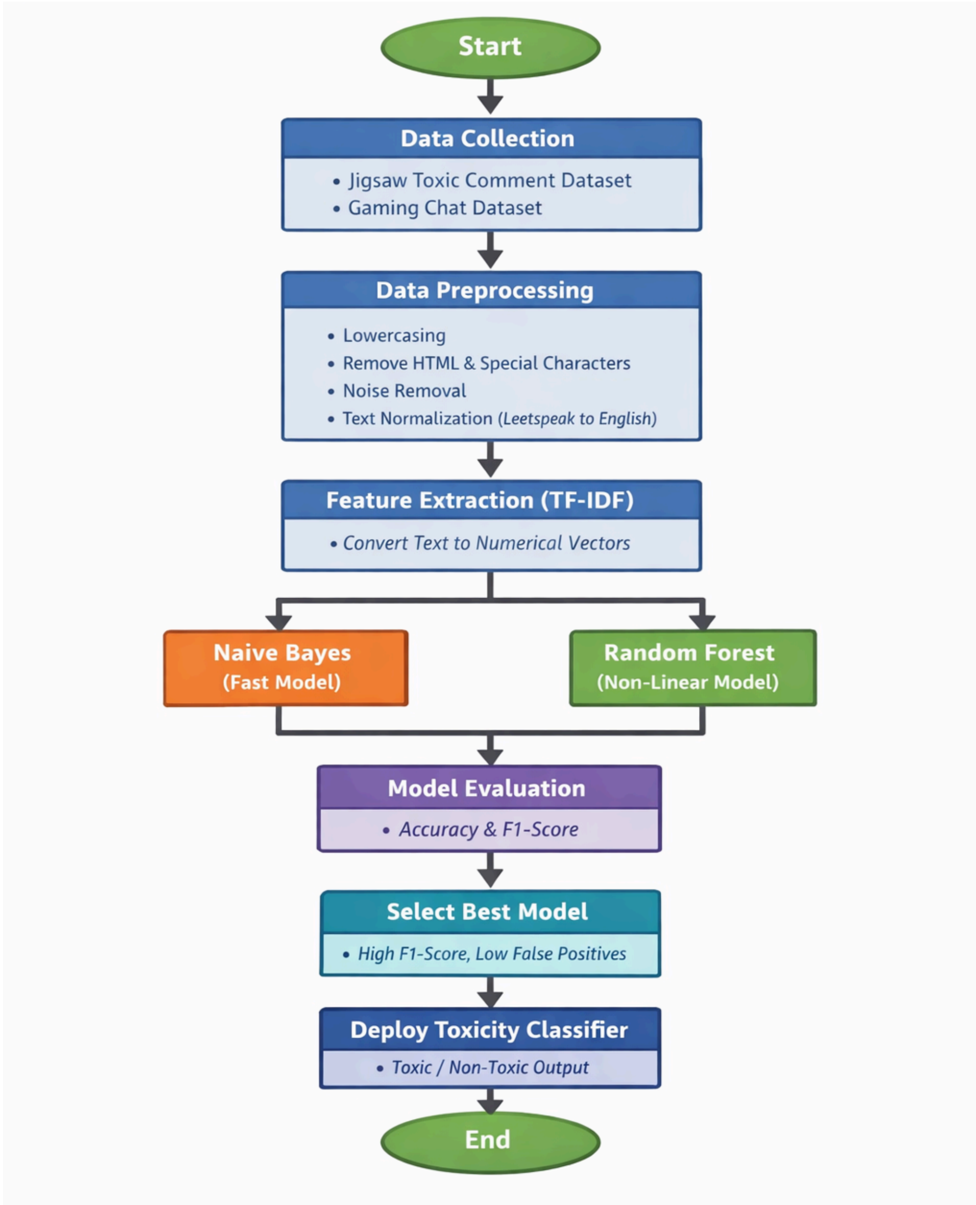
- **Environment:**

  * **Jupyter Notebook / Google Colab:** For experimental development and model training.

6. Methodology

The project will follow a standard NLP pipeline, focusing on data cleaning and feature extraction.

- **Data Collection:** We will use the Jigsaw Toxic Comment dataset or a specific Gaming Chat dataset from Kaggle/GitHub.
- **Data Preprocessing:** This includes lowercasing, removing noise (HTML, special characters), and "Normalization"—a process where we convert leetspeak (e.g., "v3ry") back into standard English.
- **Feature Extraction:** Using **TF-IDF (Term Frequency-Inverse Document Frequency)** to convert text into numerical vectors that the ML model can understand.



- **Model Selection:** We will implement and compare two beginner-friendly models:
  - **Naive Bayes:** For fast, real-time classification.
  - **Random Forest:** To handle non-linear patterns in the text.
- **Evaluation:** The model will be tested using Accuracy and F1-Score to ensure it doesn't block too many "clean" messages.

7. Conclusion

Developing a toxicity filter for kids' gaming is not just a technical challenge but a social necessity. By moving from a static list to an adaptive Machine Learning model, we can create a safer environment that understands the nuances of gamer language. This project will provide a foundation for real-time moderation that can eventually be scaled to support voice-to-text filtering and multi-language support, ensuring that digital playgrounds remain a space for positive interaction.

8. References:

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[3] H. Ismail, A. Khalil and A. Jasmy, "Enhancing Online Toxicity Detection on Gaming Networks: A Novel Embeddings-Based Valence Lexicon Approach," *Int. J. Data Sci. and Analytics*, vol. 20, pp. 4489-4500, 2025. :contentReference[oaicite:9]{index=9}

[4] "Jigsaw Toxic Comment Classification Challenge," *Kaggle Dataset* (Online). Available: https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge

Datasets:

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